Assessing LLMs' mathematical abilities requires understanding the various mechanisms of mathematical creativity

Silvère Gangloff [1]

Abstract

How should we assess whether large language models can perform mathematical invention? I argue that this question is currently underspecified: mathematical creativity is not one capacity but several mechanistically distinct modes of meaning-making - reflexive introspection on mathematical practice, analogical import from the sciences, problem-driven construction, and the bridging of distant domains - together with a further, cross-cutting distinction between meaning pursued because a pattern was observed and meaning pursued because it is strategically wanted, a distinction I develop through the case of conjecture-formation. These mechanisms are likely non-substitutable, so that competence in one does not transfer to the others. Grounding each in a historical case study and in an architecture-level account of current transformer-based systems, I suggest that today's models concentrate their competence in modes shaped by recombination and search over existing building blocks; if that description holds, the remaining modes are out of reach in principle, not just slower - though whether it holds is itself the open, empirical part. Because proof is getting cheaper as AI improves at generating it - a shift the field's own leading voices are now diagnosing - mathematical value is migrating toward the modes current systems cannot yet perform, and evaluations of AI mathematical ability should be organized around this taxonomy rather than around aggregate benchmarks that conflate it.

1. Introduction

G. H. Hardy opens *A Mathematician's Apology* by declaring that a mathematician, like a painter or a poet, is *"a maker of patterns,"* and that it is the beauty of an idea, not the rigor of its proof, that is the true measure of the work (Hardy, 1940). Yet the working life of the same Hardy tells a different story: for years, Hardy and Littlewood devoted themselves to supplying rigorous proof for claims that Ramanujan had reached by an intuition neither of them could fully reconstruct, precisely because intuition alone, however striking, was not treated by the field as sufficient grounds for trusting a claim. Hardy is therefore a witness for this paper's position even while representing the era that made proof mathematics' primary currency: proof's dominance was plausibly never because it *is* the value of mathematics, but the only scalable technology for checking an intuition. When verification becomes available by other means, the true scarce resource - whose intuition to trust, what is worth the labor of formalizing - is exposed again.

That shift is no longer speculative. In his 2026 ICM plenary, *"Mathematics in the Age of AI,"* Terence Tao argues that the field is moving from a shortage of proofs to an overwhelming abundance of them, and warns explicitly that *"the inherently ungrounded nature of generative AI, combined with the financial incentives of AI companies, makes the use of these tools particularly vulnerable to"* Goodhart's law - the tendency of a measure, once made a target, to stop being a good measure (Tao, 2026). Tao decomposes mathematical work into five stages beyond initial problem-solving: proof generation, verification, exposition, publication, and canonicalization for future generations, observing that AI is already strong at generation and improving quickly at verification, while the remaining stages still require sustained human involvement.

Let's be more precise about what this five-stage account does and does not claim, since it is easy to mistake for a competing version of the taxonomy developed here. Tao's stages are a *sequential, lifecycle* account of what happens to a piece of mathematics once it already exists as a claim: generated as a proof, checked, published, absorbed into the literature and curriculum. Every stage presupposes a claim is already in hand. This paper asks a prior question: where does the content - for instance, concepts, conjectures and axioms - come from in the first place? Tao folds that question silently into *"problem-solving,"* since his interest is

[1] IRAFM, University of Ostrava, Ostrava, Czech Republic. Correspondence to: Silvère Gangloff <silvere.f.gangloff@gmail.com>.

Preprint. August 18, 2026.

in what happens to content after it exists, not in its genesis. Turing's 1936 paper illustrates the difference: its results pass through generation, verification, and canonicalization into the computer science curriculum exactly as Tao's stages describe, but none of those stages touch the prior, reflexive move that made the paper possible - noticing that a human clerk mechanically following a rulebook could itself become a formal object (Turing, 1936). That move precedes stage one entirely. The two taxonomies are not competing slices of the same phenomenon: one classifies the lifecycle of mathematical content, the other the generative mechanism that produces it.

This paper is a direct response to an open question left by the closest existing work in this vein. Zahavy (2026) uses Peirce's classical division of inference into deduction, induction, and abduction to argue that while large language models have mastered the deductive verification of theorems and the inductive compression of data, they lack the abductive *"jump"* from sense experience to formal axioms that he takes Einstein's discovery of general relativity to exemplify, and he proposes that physically grounded, actionable world models are the missing piece. I take his argument seriously and build directly on its Peircean apparatus; I do not think it fails so much as it stops exactly where it says it does. Its conclusion explicitly, and I think correctly, declines to force a physics-derived axis onto a domain he had not examined, noting only that in mathematics *"[sense experience] may be grounded in high-dimensional topology or have other goals such as generality or minimality,"* without developing the point further. This paper departs from that single, undeveloped sentence. His account treats the missing capacity, once physics is the domain, as a single jump; I ask whether the analogous capacity in mathematics has the same unity, or whether it decomposes once mathematical practice is examined on its own terms.

My position is that it decomposes, and that treating it as a single capacity - in either direction, as uncritical optimism about AI mathematical ability or as blanket skepticism about AI creativity - is a mistake. I identify four mechanistically distinct modes, not offered as an exhaustive list, by which mathematics has historically produced new concepts, axioms, and connections: *reflexive* mathematics, which formalizes an existing mathematical or logical practice by taking the practice itself as an object of study; *analogical* mathematics, which imports structure discovered in the sciences and repurposes it inside pure mathematics; *problem-driven* mathematics, in which new concepts arise as a byproduct of solving it; and the *bridging of distant domains*, in which value comes specifically from connecting fields with no obvious prior relationship. Cutting across all four, in a way Section 2 situates precisely, is a further distinction developed through the case of conjecture-formation: a claim can be reached because a pattern was noticed and generalized, or it can instead be adopted as a target because its truth would deliver a separately wanted result, with its truth investigated only afterward. For each mode, and for this further axis, I ground the historical case in an architecture-level account of why current transformer-based systems, which operate substantially by recombining and searching over a learned library of existing moves, succeed or fail at it. My central claim is that these mechanisms are likely *non-substitutable*: strength in the modes current systems already perform well - existential search, in-space deduction, and straightforward cross-domain transfer - does not transfer to the modes that require inventing a genuinely new conceptual primitive rather than a new combination of old ones. Given recombination over a fixed candidate type, this is what a system cannot reach at all, not merely what would take longer - a completeness claim, not an efficiency one. The risk sits in that premise, not its consequence: whether current systems really are recombination engines is defended from an observed pattern in Section 3.3 onward, not proved outright.

The remainder of the paper proceeds as follows. Section 2 recalls the Peircean apparatus this paper shares with Zahavy (2026) and states the domain-specific question I ask of it. Section 3 develops the four modes and the further motivational axis in turn. Section 4 states the non-substitutability thesis directly. Section 5 argues that mathematical value is migrating toward the modes current systems cannot yet perform, as the cost of producing and verifying proofs falls with the improving performance of AI systems at generating them. Section 6 takes up credible objections to this position, and Section 7 closes by returning to the sentence in Zahavy (2026) this paper set out to answer.

2. The Peircean frame

Peirce's classical account of inference divides it according to the structural permutation of a rule, a case, and a result. Deduction (rule + case $\rightarrow$ result) applies a general rule to a particular case to derive a result with certainty; induction (case + result $\rightarrow$ rule) generalizes a rule from an accumulation of cases and their results; and abduction (rule + result $\rightarrow$ case) infers the case, or a new rule, that would best explain an otherwise surprising result (Peirce, 1934). Zahavy (2026) adopts this trichotomy to argue that large language models have mastered induction, as the statistical compression of data, and are rapidly mastering deduction, as the formal verification of theorems from given premises, but lacks a mechanism for abduction: the generation of the axioms from which deduction can proceed at all. His case study is Einstein's invention of general relativity, where the relevant axioms - chiefly the equivalence principle - were reached not by observing a body of data (little of the relevant kind yet existed) nor by deducing them from prior theory (they were themselves the premises), but by what he calls manipulative abduction: an embodied thought experiment that grounded a new axiom in simulated physical sensation. Although the case study is drawn from physics, the claim is not confined to it: his own conclusion states that *"the necessity of the Abductive Jump remains universal,"* varying only in *"the nature of the simulation,"* which must be *"adapted to the ontology of the discipline"*-for physics, the world; for mathematics, *"the abstract landscape of formal systems"* (Zahavy, 2026).

Zahavy's apparatus is a starting point, not a foundation: none of induction, deduction, or abduction alone accounts for the origin of new mathematical content, so his question - where do the new rules come from - is the right one to ask of mathematics too. What follows is not a derivation of four sub-species inside his trichotomy, but an examination of mathematical practice on its own terms, which does not support his claim that the jump is one capacity, universal in kind, varying only in substrate. New mathematical rules arise, historically, along several structurally distinct routes, triggered by different kinds of input - an existing practice to formalize, an external analogy to import, a problem's own constraints, a distant domain's latent structure. Whether each route is properly a species of Peircean abduction, or simply shares its rough shape, is left open; Sections 3.1–3.4 develop the four routes regardless, extending Zahavy's question rather than rejecting it. Section 3.5 then develops a further, orthogonal distinction - not a fifth route, but a question about what motivates the pursuit of any of the four: whether it is triggered by an observed pattern or by a separately held goal.

3. A taxonomy of mathematical meaning-making

3.1. Reflexive mathematics

Some mathematics formalizes not some external subject matter but an existing mathematical or logical practice, taking the practice itself as the object of study - what I call *reflexive* mathematics. Turing's 1936 paper is the paradigm case: it does not formalize an external phenomenon but the practice of a human *"computer"* mechanically following a fixed rulebook, and it is this reflexive move, not any subsequent theorem, that originates the Turing machine (Turing, 1936). Boole's *Laws of Thought* is an earlier and more explicit instance: Boole states directly that he is building a mathematical model of the mental operations of reasoning, not merely a notation for propositions already agreed upon, reducing connectives such as *"and," "or,"* and *"not"* to algebraic operations on truth values - the origin of the word *"Boolean"* now embedded in every programming language (Boole, 1854). Gentzen's natural deduction is a third: it supplies a pair of rules for each logical connective, one to introduce it and one to eliminate it (for instance, one rule concludes *"$A \wedge B$"* from *"A"* and *"B"* held separately, and another recovers *"A"* from *"$A \wedge B$"*), designed, in Gentzen's own words, to come as close as possible to the informal moves mathematicians already make when they argue, and its introduction/elimination structure remains the backbone of the proof systems used in modern interactive theorem provers such as Lean (Gentzen, 1935). Gödel's arithmetization of syntax, which turns the practice of proving into an object inside arithmetic itself, extends the same move to a second order (Gödel, 1931).

Whether large language models can perform this mode is not settled by asking whether they can introspect, a question on which the literature is already substantial. Fine-tuned to predict their own behavior in hypothetical scenarios, language models outperform other models at predicting them, but only on simple, in-distribution tasks (Binder et al., 2024). Concept-injection experiments, in which a vector is inserted into a model's residual stream mid-task, find that a frontier model can sometimes detect and name the injection, but unreliably (Anthropic, 2025); a direct follow-up shows that this particular binary detection paradigm is largely a methodological artifact of a global bias toward affirmative answers, while a narrower, more robust capacity survives in relative judgments - localizing which of several sentences was perturbed, and comparing the strength of two perturbations, though confined to early-layer injections and collapsing to chance thereafter (Hahami et al., 2025). Introspection, where it exists at all in these systems, is partial and layer-dependent, not a unified transparent readout.

This should not be read as a point in favor of humans by contrast. Human introspection is not a transparent readout either: people confidently report causes for their own behavior that demonstrably were not the actual causes (Nisbett & Wilson, 1977). Neither side has raw access to its own mechanism, and the disanalogy between them is subtler than it first appears. In fact, Turing's own move did not require phenomenal, first-person transparency: the object he formalized, a human clerk following a rulebook, is third-person and behaviorally observable, not introspectively private. Language models, trained on an immense record of human mathematical practice, arguably have more systematic third-person exposure to that practice than any single human mathematician does. Data access, then, is probably not the real bottleneck to reflexive mathematics.

The more likely bottleneck is *object selection*: nothing in ordinary training or inference makes an existing symbolic or cognitive practice salient as a thing worth turning into a formal object in its own right, absent someone external pointing at it. This is the structural feature shared by all four case studies above, not a peculiarity of Turing's alone: in each, what got selected for formalization was the practice itself - computing, for Turing; reasoning, for Boole; the structure of argument, for Gentzen; proof-construction, for Gödel - rather than some external subject matter the practice happened to be used on. This reframes reflexive mathematics as an instance of a more general concept-creation problem - which practice gets selected as worth formalizing - rather than a separately blocked introspective faculty. One candidate account of how such selection happens is retrospective: a candidate formalization is proposed, and its fruitfulness, how much rich, unifying structure follows from it, validates the choice after the fact. If this is right, object selection resembles a search-and-score loop, propose a formalization and evaluate it by downstream theorem-yield, much closer to what evolutionary and search-based AI systems already do than to a mysterious human faculty, and it directly extends Zahavy's own suggestion that mathematics' sense experience may be grounded in *"generality or minimality."* This is a genuine partial rescue, but I do not think it closes the gap, because it addresses only half the problem. Fruitfulness-scoring evaluates a candidate once one exists; it says nothing about how the candidate's *type* arises in the first place. Systems that propose and score candidate mathematical objects, such as AlphaEvolve and FunSearch, search a space whose type is already fixed by the problem statement - find a set, a sequence, a construction of an already-named kind (Novikov et al., 2025; Romera-Paredes et al., 2024). Turing's case makes this clearest: he did not search a pre-given space of *"possible formalizations of computation"* and score its members; he first had to treat *"the practice of a human calculating"* as a legitimate kind of object to formalize, before any scoring loop could begin, and the same holds for Boole's choice of reasoning, Gentzen's choice of argument, and Gödel's choice of proof as objects worth formalizing in their own right. The scoring loop is the easy part; picking what to run it over is what reflexive mathematics actually demands, and that is where current systems stall.

3.2. Analogical mathematics

A second mode creates new concepts by importing structure discovered in the sciences into pure mathematics, where it is repurposed to answer questions with no dependence on the domain it came from; I call it *analogical* mathematics. The physical sciences supply the clearest running examples below, but nothing in the mechanism restricts the source to physics specifically: biology, for instance, is just as plausible a source domain in principle. A familiar instance of the move is Fourier's decomposition of heat conduction into sinusoidal modes, introduced to solve a specific physical problem and subsequently generalized into harmonic analysis, a foundational area of pure mathematics with no remaining dependence on heat or on any other physical process. The clearest *sustained* example, with the longest lineage of major results built on the imported concept, is entropy. Ergodic theory had, by the 1950s, an unsolved classification problem: given two Bernoulli shifts, decide whether they are measure-theoretically isomorphic, with no known invariant able to tell most pairs apart. Kolmogorov and Sinai's move was to notice that the statistical entropy of Boltzmann and Gibbs, redefined appropriately, behaves as exactly such an invariant for abstract measure-preserving dynamical systems (Kolmogorov, 1958; 1959; Sinai, 1959). Ornstein closed the loop in 1970, proving the invariant complete: two Bernoulli shifts are isomorphic exactly when their Kolmogorov–Sinai entropies agree (Ornstein, 1970). Decades later the same invariant was still doing new work: Bowen extended it to actions of countable sofic groups, resolving isomorphism questions the classical invariant could not reach (Bowen, 2010), showing that the imported concept is still generative decades after the original analogy was drawn.

Of the four modes in this taxonomy, this is the one I am most conditionally optimistic about. Zahavy (2026) argues that physically grounded, actionable world models - systems that support intervention on a simulation, such as Genie (Bruce et al., 2024), rather than merely predicting its next frame - could supply the sensory grounding current language models lack. We can extend this specifically to analogical mathematics: a world model that has extracted genuine physical understanding, not merely pixel-level prediction, could plausibly extract mathematical objects like entropy from that understanding and repurpose them for questions with no physical content. Concretely: repeated counterfactual intervention on a physical system - turning a dial, running the same process forward from different initial conditions - exposes some quantity as invariant, or as changing only in one direction, independently of the specific physical instantiation being manipulated. Some such pattern of invariance becoming *salient* as a candidate for abstraction is plausibly what let Kolmogorov and Sinai treat a physical quantity as the right analogy for an abstract isomorphism invariant, whatever the specific route by which they themselves arrived at it; reproducing that kind of salience, not grounding in general, is what a world model would need to support for analogical mathematics to become reachable by an AI system.

3.3. Problem-driven mathematics

A third source of new concepts is neither a formalized practice nor an imported structure, but a specific problem solved, with the concepts arising only as a byproduct: *problem-driven* mathematics. This mode splits along a line that matters for what current systems can do with it. In its *existential* or constructive form, the goal is a witness: a counterexample, or a construction meeting some extremal constraint. A great deal of what current AI-assisted mathematics has actually produced is of this shape: FunSearch and AlphaEvolve's headline results - improved constructions for the cap set problem, better bounds on kissing numbers, faster matrix-multiplication algorithms - are almost uniformly existential (Romera-Paredes et al., 2024; Novikov et al., 2025). In its *universal* or structural form, by contrast, the goal is a new organizing theory that reorganizes an entire domain rather than answering one existential question, and no accumulation of witnesses could have produced it. Abel had shown by 1824 that no general formula in radicals solves the quintic, but this left open why some specific quintics are solvable and others are not - a question no list of examples could settle. Galois's answer was to associate to each polynomial a group of permutations of its roots and to show that solvability by radicals corresponds to a specific structural property of that group. The result was not a new technique for solving equations but a new kind of object, group theory itself, under which the whole question could be redescribed and closed. Non-Euclidean geometry answers a structurally similar resistance - centuries of failed attempts to prove the parallel postulate from the other four - with a comparable move: a new kind of object, a consistent geometry in which the postulate simply fails, rather than a witness. Both cases are shaped like the abductive jump of Zahavy (2026): a persistent, surprising resistance resolved not by deduction or generalization, but by inferring an entirely new explanatory apparatus.

The architectural story developed throughout this taxonomy is a conjecture inferred from an observed pattern, not an established mechanistic fact. Essentially every headline result from autonomous, self-directed AI mathematical discovery - where the system chooses what to search for, not merely completes a proof already posed - is existential; separately, a mechanistic study of language models solving arithmetic finds them relying on a sparse set of narrow, interpretable heuristics rather than a general procedure (Nikankin et al., 2025), indirect evidence, at a much smaller scale, for the same picture. Taking the pattern at face value: a transformer's attention mechanism, refined by search or reinforcement learning over candidate proof steps, plausibly recombines a learned repertoire of existing moves. This repertoire need not be formalized: alongside named lemmas, tactics, and constructions already seen in training or discovered by search, it includes looser fragments of informal reasoning that a human reader would not necessarily isolate as a standalone step within the surrounding proof. Recombination, whether over formal or informal moves, is naturally shaped like witness-finding: search a combinatorial space of known gadgets for one instance satisfying a constraint, exactly what an existential claim, or a counterexample to a universal one, asks for. It is not naturally shaped like structural reorganization, which asks not for an instance of a known kind of object but for a new kind of object under which the whole domain can be redescribed.

What makes a candidate object genuinely new, rather than a fresh assembly of existing pieces, is not something this paper resolves: every new definition is, at some level, built from existing primitives. One candidate criterion, offered as a position rather than a proof: a new counterexample is typically long and specific to the instance that produced it; a new concept is typically short, and opens results not provable before it existed. The criterion is retrospective, not a test a system's output could pass on the spot - newness becomes clear only once downstream consequences do - but it still predicts something: any AI-discovered concept later confirmed genuine should turn out short and reused, not a longer witness mistaken for one. Gowers (2026) independently draws the same line from live experience with frontier systems: search breadth beats any mathematician, pruning judgment does not, and genuine progress is measured by an equally retrospective standard - a technique unremarkable in hindsight but *"difficult to stumble on by accident."* His observations stay entirely inside problem-driven mathematics, though, and say nothing about whether the same line holds for reflexive formalization or bridging distant domains - the generality claim this paper makes and his does not. That Zahavy (2026) and Gowers (2026) converge on the same split from opposite directions - philosophy of mind and live experimentation - is evidence it tracks something real, not an artifact of either paper's framing.

3.4. Bridging distant domains

A fourth mode's value comes specifically from connecting two bodies of mathematics - entire fields, but just as often two specific objects, theorems, or results within or across them - with no obvious prior relationship: the *bridging* of distant domains, historically one of the most prized forms of mathematical unification. Part of why it is prized is a matter of division of labor: mastering one field well enough to contribute to it already absorbs most of a working mathematician's career, so mastering several fields well enough to notice that a bridge between them is even possible is correspondingly rarer, and unifying results have carried outsized prestige in part because so few people are ever positioned to attempt them. This constraint does not obviously bind an AI system trained on the literature of many fields at once, unbottlenecked by a single lifetime of specialization - one reason, developed below, that surfacing a candidate connection is plausibly closer to reach than constructing one where none yet exists. This mode also splits usefully into two cases. In the first, a translation dictionary between the two domains already exists, or becomes available once the right quantities are computed on both sides; connecting them is then a matter of recognizing an already-available plug. In the second, no such interface exists, and bridging the domains requires inventing new intermediate machinery from scratch. Davies et al. (2021) give a clean, concrete instance of the difference between finding a plug and filling a gap. A supervised model trained to relate algebraic invariants of knots to their hyperbolic-geometry invariants (and, separately, a structure connecting representation theory and combinatorics) surfaced a strong statistical correlation between quantities already computable on both sides of each divide - exactly the *"plug"* case. But it took a human mathematician to construct the theorem explaining why the correlation held, the actual bridging move in the sense that matters for the harder case. Current systems look plausibly capable of the first case and structurally unequipped for the second, for the same architectural reason given throughout this section: recombining what is already computable is a search problem; building the interface where none exists is a concept-creation problem.

A related mechanism worth distinguishing from bridging, though space precludes developing it as its own case here, is *unification*: rather than an interface between two existing structures that remain standing on either side of it, unification defines a single, more general object into which several existing objects particularize as special cases, sometimes from domains with no prior relationship at all - group theory's absorption of the many symmetry groups already in separate use, or category theory's later absorption of structures across most of mathematics, are paradigm instances. Where bridging adds a connection, unification typically dissolves the connected structures into instances of something new, which can have the further effect, when it reaches across domains, of transferring techniques from one instance to all the others. Whether this is truly independent of the constructive move behind Sections 3.3–3.5, or another guise of it, is left to future work.

3.5. Observation-driven versus goal-driven conjecture

As anticipated in Section 2, the axis this section develops is sharpest in the case of conjecture-formation: a conjecture is *observation-driven* when it is reached inductively, by noticing a pattern across many cases and generalizing it, and *goal-driven* when a claim is instead adopted as a target because, if true, it would deliver a separately wanted result, with its truth investigated only afterward. The Collatz conjecture is observation-driven in exactly this sense: it generalizes from an accumulation of computed trajectories, with no external motivating goal beyond the pattern itself. The route to Fermat's Last Theorem through the Frey curve is goal-driven in the contrasting sense. Frey proposed that a specific elliptic curve, built from a hypothetical counterexample to Fermat's Last Theorem, would fail to be modular (Frey, 1986); Ribet proved the corresponding epsilon conjecture, confirming that this failure would indeed follow (Ribet, 1990); and Wiles, together with Taylor, proved enough of the modularity conjecture to complete the theorem (Wiles, 1995; Taylor & Wiles, 1995). The entire program of proving modularity for this specific curve was pursued because it would deliver Fermat's Last Theorem, not because of independent pattern-noticing about elliptic curves - a goal-driven conjecture in a sense Collatz never was.

This axis cuts across the same depth distinction developed in Sections 3.1–3.4, rather than resolving to a single verdict. In its shallow, in-space form, goal-driven reasoning is goal-directed subgoal search within an already specified space of lemmas and constructions - essentially backward chaining: work back from the desired conclusion, propose an intermediate lemma, and check whether existing tactics close the resulting subgoal. This is the same recombination mechanism described throughout this taxonomy, applied under an explicit goal rather than free exploration, which is why it is close to what reinforcement-learning-guided theorem provers, already searching over exactly this kind of fixed space via expert iteration, do reasonably well. In its deep, out-of-space form, exemplified by Frey's move, the target claim requires constructing a genuinely new mathematical object as the connecting device - a specific elliptic curve built from a hypothetical counterexample, whose relevance had no precedent to search over - not selecting among known lemmas, before its truth is even known. Nothing in the space of previously considered elliptic curves would have flagged this particular construction as worth proposing; the construction itself, and not merely its eventual truth, was the creative act. The first case looks reachable by current search-augmented systems because it searches an already-fixed candidate type, exactly the sense in which Section 3.1 found the evaluation half of reflexive mathematics tractable; the second is stuck for the same reason every hard case in this taxonomy is stuck - there was no space of candidates to search until Frey built one.

4. Non-substitutability of mechanisms

The four modes developed in Section 3, together with the further motivational axis developed alongside them, are not independent paths to the same destination. My position is that they are distinct routes into what a mathematician would recognize as concept space, and that they are probably not mutually substitutable: strength in one does not transfer to the others. The claim is about reachability, in other words, rather than speed: certain mathematics sits outside what the architecture current systems share can produce at all, regardless of how much longer a slower search might be given.

The same underlying mechanism explains both the pattern of successes and the pattern of gaps across every section above. Attention-based recombination over a learned library of existing moves, refined by search or reinforcement learning, is well suited to tasks whose candidate space has a fixed, already-specified type: deduction from given axioms, the existential half of problem-driven mathematics (Section 3.3), the plug half of bridging distant domains (Section 3.4), and the shallow half of goal-driven conjecture (Section 3.5). It is not well suited to tasks that require inventing the candidate type itself, rather than searching within one already given: reflexive mathematics (Section 3.1), the structural half of problem-driven mathematics, the gap-filling half of bridging distant domains, and the deep half of goal-driven conjecture. Analogical mathematics (Section 3.2) sits apart from this pattern precisely because it is conditional on a different kind of architectural progress, grounded world models, rather than on scaling the recombination mechanism that governs the rest of the taxonomy. Concept creation, in the sense of establishing a new stable primitive rather than a new combination of existing ones - close to Boden (2004)'s transformational creativity, against the combinational and exploratory creativity recombination already covers - is therefore not one item on a longer list of missing capabilities; it is the single recurring bottleneck that the taxonomy in Section 3 decomposes into four historically distinguishable mechanisms, cut across in every case by the further question of what motivated their pursuit. Borrowing the vocabulary from Section 2 without depending on it: every intractable half has the rough shape of abduction, inferring something new rather than applying or generalizing what is given; every tractable half has the rough shape of deduction or induction over an already-fixed space. The claim survives without the vocabulary: one missing capacity in a single physics case study recurs, distributed across mathematical practice, as several distinct obstacles with a family resemblance.

Whether the structural half of problem-driven mathematics, the gap-filling half of bridging, and the deep half of goal-driven conjecture are genuinely separate mechanisms, or one constructive move under three different triggers, is a question the historical cases alone probably cannot settle, and this paper does not try to. That caps how strong the taxonomic claim can be; what survives the cap is narrower, but real: a resistant problem, an interface-less pair of domains, and a goal with no existing lemma are recognizably different situations to notice, whatever their deeper relationship turns out to be. Organizing around what must be perceived to trigger a construction, rather than around its deepest unity, is what makes the evaluation proposal in Section 5 useful either way - and why bridging keeps its own section regardless of how that further question comes out.

5. Where mathematical value is migrating

Section 1 opened with Tao's diagnosis that the field is moving from a shortage of proofs to an overwhelming abundance of them (Tao, 2026). If Section 4 is right that this abundance is concentrated in modes built from recombination and search - deduction, existential construction, straightforward transfer - rather than in modes that require new conceptual primitives, then the practical upshot is not merely that the cost of proof is falling in general. It is that mathematical value, both the comparative advantage of individual mathematicians and the allocation of the field's collective effort, should migrate toward precisely the modes Section 3 identifies as currently out of reach: reflexive formalization of practice, analogical import pending grounded world models, structural reorganization, gap-filling between domains, and the deep, goal-driven form of conjecture.

This migration is not unambiguously good news; it runs in two directions at once. On one branch, less costly proof frees mathematicians from the labor of verification to spend more of their attention on exactly the concept-level work this taxonomy identifies as still scarce. On the other, Tao's own warning is that *"the inherently ungrounded nature of generative AI, combined with the financial incentives of AI companies, makes the use of these tools particularly vulnerable to"* Goodhart's law (Tao, 2026): once a measure such as proof count or benchmark performance becomes a target, it stops being a good measure. A field that already produces proofs of uninteresting results because proof itself is automatically valued, independent of whether the result is interesting, is a dynamic visible before any of the recent AI systems existed; driving the cost of proof down further could just as easily flood the literature with more of the same rather than freeing anyone's attention for anything else. Both branches follow from the same premise - that recombination-shaped work is becoming abundant - and which one dominates in practice depends on incentives this paper does not attempt to redesign, only to name.

This is a recommendation, not only a diagnosis. If AI mathematical systems' value is concentrated in a specific subset of Section 3's modes, a single aggregate score for *"mathematical ability"* measures the wrong thing: it cannot distinguish strength at existential search from a genuinely new mode of concept creation, and under Goodhart's law an aggregate invites optimizing the number rather than the capacity itself. Benchmark designers and evaluators should report performance broken out by mode, rather than folding reflexive, analogical, problem-driven, and bridging tasks, at any point on the observation/goal axis, into one figure - making the taxonomy falsifiable by disaggregation, not just defensible as philosophy.

6. Alternative views

A credible alternative holds that recombination is not fundamentally limited, only under-scaled: enough compute, better search scaffolding, or expert-iteration training like AlphaProof's might reach the modes called structurally out of reach here. But scaling recombination searches a candidate type more thoroughly; it does not expand what counts as one, the claim in Section 4 - a difference in kind, not degree. This needs a case where scaled search alone produced a genuinely new primitive rather than a better search of an old one. None exists that I know of, but I hold the claim as a bet that could lose, not a matter of principle. A related version is that prompting might already elicit the missing capacity, a concern raised publicly against the closest prior work in this vein[1]: prompting works within an existing symbolic space, combining what is already there, which is why it does not reach outside it.

The next objection runs against my optimism rather than my pessimism: Section 5 argued cheaper proof could free mathematicians for concept-level work, but conceded, following Tao, that it could just as easily degrade the field's signal-to-noise with more correct, uninteresting results. I regard this as a live, unresolved tension, not something the taxonomy resolves.

Section 3.1 faces another objection: if fruitfulness-scoring rescues the evaluation half of object selection, perhaps it rescues generation too, undermining the claim that reflexive mathematics stays out of reach. Doubtful, because every search-and-score system known searches a space whose type the problem statement already fixed; no fruitfulness criterion tells a system which space to search in the first place. This is an empirical bet, not a first principle, and it loses the moment a system proposes a new category of object, not just a new member of an old one, later vindicated by its consequences.

The diagnosis in Section 3.1 that data access is not the bottleneck to reflexive mathematics is questioned too: language models are trained on an enormous third-person record of human mathematical practice, so perhaps that exposure already suffices, and object selection is an unnecessary further bottleneck. My response is that exposure to a practice and treating it as a salient object of formalization are different achievements: a system trained on transcripts of computation does not thereby select computation itself, rather than any other regularity in its training data, as worth axiomatizing. The disagreement is about salience, not sufficiency, and I think the distinction survives.

One last objection concerns significance, not truth: that this taxonomy merely illustrates Zahavy's abductive jump in a new domain. It corrects his claim rather than instantiating it. His conclusion asserts that *"the necessity of the Abductive Jump remains universal"*; the position here is that this holds only at the level of Peirce's most general category, and is radically underspecified at the level that matters for building or evaluating a system, since what triggers the inference differs by mechanism - a practice made salient, a structure made available through grounded simulation, or a resistant problem, missing interface, or unreachable goal made salient as such. Moving from a name for the missing capacity to a falsifiable account of what must be supplied, case by case, is what a single case study could not have produced.

[1] Reviews of the closest prior work in this vein post publicly on OpenReview post-decision; this concern appears in a review of Zahavy (2026).

7. Conclusion

This paper asked how mathematical creativity should be assessed once large language models are capable of producing correct, verified mathematics at scale, and argued that the question is currently underspecified. Mathematical concept-creation decomposes into four mechanistically distinct modes - reflexive, analogical, problem-driven, and the bridging of distant domains - cut across by a further axis distinguishing conjecture reached from observation and conjecture pursued out of a separately held goal, and I have argued that these mechanisms are probably non-substitutable, so that a system's competence in the modes shaped by recombination and search does not license any inference about its competence in the modes that require inventing a new conceptual primitive. This is why I think blanket claims about AI and mathematical ability, in either direction, are mistaken: the right question is never simply whether a system can do mathematics, but which of these mechanisms a given claim of capability or incapacity is actually about.

This paper began from a single sentence left open by Zahavy (2026), who correctly declined to force his physics-derived account of the abductive jump onto a domain he had not examined, suggesting only that mathematics' sense experience *"may be grounded in high-dimensional topology or have other goals such as generality or minimality."* I have tried to take that sentence seriously rather than merely cite it: not by proposing a second jump to sit beside his, but by asking whether the very idea of a single jump survives contact with how mathematics has actually produced new axioms, concepts, and conjectures. My answer is that it does not, and that the taxonomy developed here is one way, not the only one, of taking his open question further.

Two threads are left underdeveloped for space, not substance: operationalizing object selection as the bottleneck to reflexive mathematics (Section 3.1); and proof's rise as mathematics' currency, sketched briefly through Hardy and Ramanujan. Neither changes the position argued; the taxonomy itself is incomplete, one contribution among others.

References

Anthropic. Emergent introspective awareness in large language models. https:

//transformer-circuits.pub/2025/introspection/index.html, 2025.

Binder, F. J., Chua, J., Korbak, T., Balesni, M., Rager, C., Kori, C., Meek, N., and Evans, O. Looking inward: Language models can learn about themselves by introspection. *arXiv preprint arXiv:2410.13787*, 2024.

Boden, M. A. *The Creative Mind: Myths and Mechanisms*. Routledge, 2nd edition, 2004.

Boole, G. *An Investigation of the Laws of Thought, on Which Are Founded the Mathematical Theories of Logic and Probabilities*. Macmillan and Co., London, 1854.

Bowen, L. Measure conjugacy invariants for actions of countable sofic groups. *Journal of the American Mathematical Society*, 23(1):217–245, 2010.

Bruce, J., Dennis, M., Edwards, A., Parker-Holder, J., Shi, Y., Hughes, E., Lai, M., Mavalankar, A., Steigerwald, R., Apps, C., Aytar, Y., Bechtle, S., Behbahani, F., Chan, S., Heess, N., Gonzalez, L., Osindero, S., Ozair, S., Reed, S., Zhang, J., Zolna, K., Clune, J., de Freitas, N., Singh, S., and Rocktäschel, T. Genie: Generative interactive environments. arXiv:2402.15391, 2024.

Davies, A., Veličković, P., Buesing, L., Blackwell, S., Zheng, D., Tomašev, N., Tanburn, R., Battaglia, P., Blundell, C., Juhász, A., Lackenby, M., Williamson, G., Hassabis, D., and Kohli, P. Advancing mathematics by guiding human intuition with AI. *Nature*, 600:70–74, 2021.

Frey, G. Links between stable elliptic curves and certain Diophantine equations. *Annales Universitatis Saraviensis. Series Mathematicae*, 1:1–40, 1986.

Gentzen, G. Untersuchungen über das logische Schließen. I, II. *Mathematische Zeitschrift*, 39:176–210, 405–431, 1935.

Gödel, K. Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I. *Monatshefte für Mathematik und Physik*, 38:173–198, 1931.

Gowers, T. What sort of maths are LLMs good at? Blog post, https://gowers.wordpress.com/2026/08/12/what-sort-of-maths-are-llms-good-at/, 2026.

Hahami, E., Sinha, I., Jain, L., Kaplan, J., and Hahami, J. Detecting the disturbance: A nuanced view of introspective abilities in LLMs. *arXiv preprint arXiv:2512.12411*, 2025.

Hardy, G. H. *A Mathematician's Apology*. Cambridge University Press, 1940.

Kolmogorov, A. N. New metric invariant of transitive dynamical systems and endomorphisms of Lebesgue spaces. *Doklady Akademii Nauk SSSR*, 119(5):861–864, 1958.

Kolmogorov, A. N. Entropy per unit time as a metric invariant of automorphisms. *Doklady Akademii Nauk SSSR*, 124:754–755, 1959.

Nikankin, Y., Reusch, A., Mueller, A., and Belinkov, Y. Arithmetic without algorithms: Language models solve math with a bag of heuristics. In *International Conference on Learning Representations (ICLR)*, 2025. arXiv:2410.21272.

Nisbett, R. E. and Wilson, T. D. Telling more than we can know: Verbal reports on mental processes. *Psychological Review*, 84(3):231–259, 1977.

Novikov, A., Vũ, N., Eisenberger, M., Dupont, E., Huang, P.-S., Wagner, A. Z., Shirobokov, S., Kozlovskii, B., Ruiz, F. J. R., Mehrabian, A., Kumar, M. P., See, A., Chaudhuri, S., Holland, G., Davies, A., Nowozin, S., Kohli, P., and Balog, M. AlphaEvolve: A coding agent for scientific and algorithmic discovery. arXiv:2506.13131, 2025.

Ornstein, D. S. Bernoulli shifts with the same entropy are isomorphic. *Advances in Mathematics*, 4(3):337–352, 1970.

Peirce, C. S. *Collected Papers of Charles Sanders Peirce*. Harvard University Press, 1934.

Ribet, K. A. On modular representations of $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ arising from modular forms. *Inventiones Mathematicae*, 100 (2):431–476, 1990.

Romera-Paredes, B., Barekatain, M., Novikov, A., Balog, M., Kumar, M. P., Dupont, E., Ruiz, F. J. R., Ellenberg, J. S., Wang, P., Fawzi, O., Kohli, P., and Fawzi, A. Mathematical discoveries from program search with large language models. *Nature*, 625(7995):468–475, 2024.

Sinai, Y. G. On the notion of entropy of a dynamical system. *Doklady Akademii Nauk SSSR*, 124:768–771, 1959.

Tao, T. Mathematics in the age of AI. ICM 2026 plenary lecture, https://teorth.github.io/tao-web/slides/age-of-ai-icm-2026.pdf, 2026.

Taylor, R. and Wiles, A. Ring-theoretic properties of certain Hecke algebras. *Annals of Mathematics*, 141(3):553–572, 1995.

Turing, A. M. On computable numbers, with an application to the Entscheidungsproblem. *Proceedings of the London Mathematical Society*, 42:230–265, 1936.

Wiles, A. Modular elliptic curves and Fermat's last theorem. *Annals of Mathematics*, 141(3):443–551, 1995.

Zahavy, T. Position: LLMs can't jump. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026.